

Project Design Phase

Solution Architecture

Date	15 February 2025
Team ID	
Project Name	AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

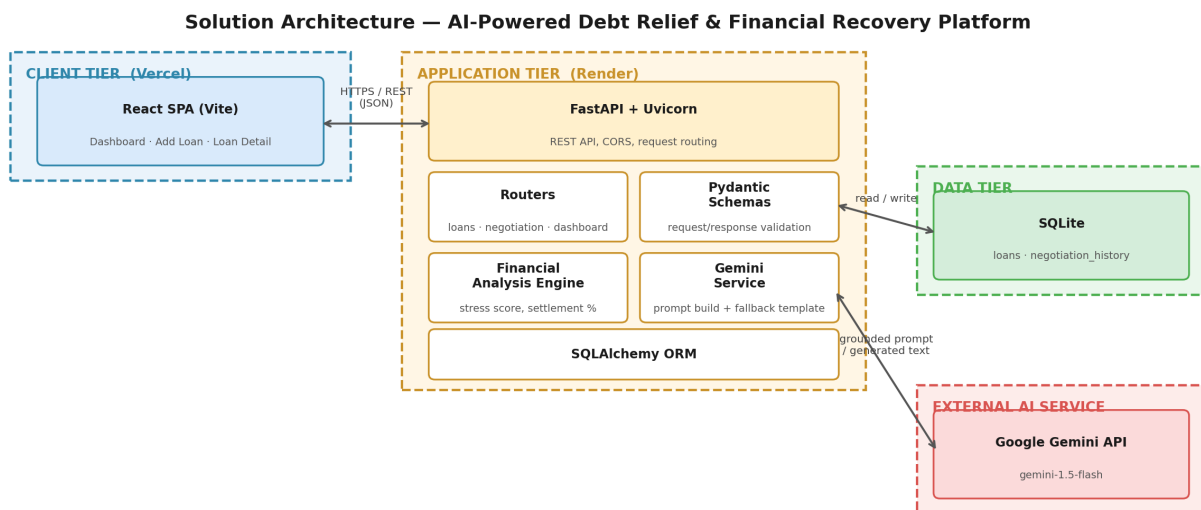


Figure 1: Architecture and data flow of the AI-Powered Debt Relief & Financial Recovery Platform

Architecture Summary:

- Client Tier: a React (Vite) single-page application hosted on Vercel, covering the Dashboard, Add Loan, and Loan Detail pages, communicating with the backend over HTTPS/REST in JSON.
- Application Tier: a FastAPI service hosted on Render. Routers handle loans, negotiation, and dashboard endpoints; Pydantic schemas validate every request; the Financial Analysis Engine computes debt-stress scores and settlement recommendations; the Gemini Service builds grounded prompts (with a template fallback); SQLAlchemy provides the ORM layer.
- Data Tier: a SQLite database storing the loans and negotiation_history tables, auto-created on first run.
- External AI Service: the Google Gemini API (gemini-1.5-flash), called only for natural-language generation of negotiation letters — never for financial calculations.